Year:	**2011**
Project Title:	Grapheme-colour synaesthesia may not improve performance in tests of spelling ability.
Name:	**Suzy Percy**

DECLARATION

I herby declare that this is an original piece of empirical research conducted and the work contained herein is entirely my own. This research complies with the ethical guidelines of the Psychological Society of Ireland and has been fully approved by the Research Ethics Committee of the School of Psychology in Trinity College, Dublin (see Appendix I for letter of approval). All data has been coded and will be stored in a secure file for a period of ten years in line with the data storage policy of Trinity College, Dublin.

Signed: _____ Date: _____

Word Count: 5,997

Introduction Word Count: 1,904

ACKNOWLEDGEMENTS

I must say a huge thank you to my dedicated supervisor, Prof. Fiona Newell, for her enthusiasm, her reassurance and for each rapid reply to my many emailed questions during the process – your input was invaluable. Thank you to Fearghal O'Brien, for his endless patience and good humour while helping me to program the Superlab experiments. I want to thank my friend Ayokunle Soyingbe, who spent the best part of a day helping me organise the vast reaction time data in Excel. Most importantly, thank you to God for his endless provision of energy, peace and helpers along the way.

ABSTRACT

Grapheme-colour synaesthesia is a familial condition in which letters, numbers, days of the week and months of the year elicit colours overlaid on a printed page or in the mind's eye. The present study implemented novel methodologies to explore memory and visual search faculties (both involved in spelling) in nine grapheme-colour synaesthetes and thirty-two non-synaesthete controls from an undergraduate population, aiming to address contradictory findings in the synaesthesia literature. Participants completed a word recognition task following the learning of 27 non-words, and a letter search task to identify target letters amongst common English words. A series of Mann-Whitney U tests found no significant differences between the groups for speed or percentage accuracy in either task, flouting expectations that synaesthesia would provided distinctive colour markers to aid performance. Correlational analyses did not reveal any relationship between the number of unique synaesthetic colours and scores in either task; nor did a majority of synaesthetes report the active use of these colours to aid memory. These findings indicate that synaesthetic colours may not provide any significant advantage in spelling ability, either in vocabulary learning or in spelling accuracy.

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INTRODUCTION

Synaesthesia is a familial, perceptual condition in which sensory stimuli elicit unusual secondary experiences in the same or a different modality (Cytowic, 1989). For example, spoken words or sounds would elicit a taste (Ward & Simner, 2003), or a smell would cause the sensation of a colour in the mind's eye (Cytowic, 1993). Grapheme-colour is most common form of synaesthesia (estimated prevalence between 1 in 2000 [Cytowic, 1989] and 1 in 25,000 [Baron-Cohen, Burt, Smith-Laittan, Harrison, & Bolton, 1996]), in which letters, digits, words, days of the week and months of the year elicit colours in the mind's eye or overlaid on the text while reading (Ward, Li, Salih & Sagiv, 2007; Simner, 2006). The present study sought to investigate the influence of grapheme-colour synaesthesia on spelling ability, the findings of which will have huge implications for methodology design in cognitive psychology, linguistics and neuroscience research where word stimuli are used. There are similar implications for teaching practices in educational contexts, where a minority of individuals who perceive additional colours may process word stimuli in a radically different way.

Grapheme-colour synaesthetes are thought to use their letter or word colour assignments (whether intentionally or implicitly) to act as distinctive markers to differentiate target characters or to act as cues for later recall; thus serving to make to-be-learned material more memorable (Cytowic, 1989; Rich, Bradshaw, & Mattingley, 2005). Classic theories suggest that the uniqueness of encodings during learning may provide memory traces that can be more readily discriminated from other traces and thus retrieved more efficiently, such as in Tulving's theory of encoding specificity (Tulving, 1979) or in Craik and Lockhart's levels of processing theory (Moscovitch & Craik, 1976). Findings show that verbal information may be better retained if paired with a visual image (Anderson & Bower, 1973; Paivio, 1969) and that imagineability significantly enhances recall (Christian, Bickley, Tarka, & Clayton, 1978).

Some research has found a significant advantage for synaesthetes in tests of verbal ability compared to matched controls (Cytowic, 2002; Mills, Innis,

Westendorf, Owisianecki & McDonald, 2006; Yaro & Ward, 2006), whereas others have reported only a slight advantage, with scores remaining within normal limits (Rothen & Meier, 2010) or no advantage at all accredited to synaesthetic colours (Rothen & Meier, 2009). While studies have assessed verbal ability using auditory learning tasks (Mills et al., 2006) and memory capacity using digit span tasks (Rothen & Meier, 2010), and the recall of digit matrices (Yaro & Ward, 2007), there has yet to be a study which involves the learning and recalling of completely novel words in a simple spelling task amongst a synaesthetic population.

In response to the present conflict in the literature and considering the probable influence of synaesthetic experiences on the learning and recall of spellings, the present study aimed to implement novel testing methodology to address the current ambiguity with regard to grapheme-specific memory encoding and word recognition. It was expected that grapheme-colour synaesthetes would more quickly and accurately identify the correct or incorrect spellings of previously learned non-words and the presence of a target letter in a series of real words, compared to controls.

In studies of spelling, the role of word or letter recognition may indicate key stages of processing that occur during letter order assembly in word formation (Falikman, 2010; Noice & Hock, 1987). Synaesthesia research has provided conflicting evidence for the role of colour concurrents in letter or word recognition (Sagiv & Robertson, 2005; Edquist et al., 2006). While some research has provided evidence indicating that grapheme-colour synaesthetes are faster and more efficient at locating specific targets during visual search tasks [in which a target character is to be found amongst a number of achromatic distracters] (Ramachandran and Hubbard, 2001a; Palmeri et al., 2002; Laeng et al., 2004), others have found no difference in performance when compared with controls (Edquist, Rich, Brinkman, & Mattingley, 2006; Sagiv, Heer, & Robertson, 2006).

There is some evidence to suggest that targets defined by a unique synaesthetic colour may attract attention during a serial scan across the

distracters (Smilek, Dixon, & Merikle, 2003; Smilek, Dixon, Cudahy, & Merikle, 2001), or increase the efficiency of distracter rejection (Palmeri, Blake, Marois, Flanery, & Whetsell, 2002), implying that the locus of synaesthetic induction occurs early in perceptual processing. Some argue focused attention is a critical factor for the perception of synaesthetic colours as colours seemed to disappear when their attention was engaged elsewhere (Mattingley, Payne and Rich 2006; Rich and Mattingley, 2003). However, other findings suggest that synaesthetic colours do not occur early enough in perception to attract or guide attention to the target (Laeng, Svartdal, & Oelmann, 2004; Sagiv & Robertson, 2005; Robertson, 2003). It must be considered however, that the classic visual search paradigm may lack ecological validity as letters are not often presented in a scattered array in everyday grapheme processing, but rather embedded in words or as single letters in an upright position. Research has shown that letters are more rapidly identified when part of a word rather than when the letter is presented alone, termed the 'word superiority effect' (McClelland & Rumelhart, 1981; Reicher, 1969), therefore a letter search task using common words of the vernacular is supported as a more ecologically valid experimental method.

The existing research highlights a number of factors that must be considered when examining the different elements of spelling amongst grapheme-colour synaesthetes. The literature describes exceptional memory abilities in a number of case studies (Luria, 1968; Smilek, Dixon, Cudahy, & Merikle, 2002), but concerns have been raised as these individuals came to the attention of the researchers due to their extraordinary abilities (Rothen & Meier, 2010). For example, savant Solomon Shereshevsky, revealed tremendous talent in memorizing huge number matrices, complex mathematical formulas, and poems written in foreign languages in just a few minutes (Luria, 1968), and claimed that his extraordinary memory span was at least in part due to his multiple synaesthesias. Smilek and colleagues (2002) similarly discovered synaesthete 'C' when she reported almost perfect scores on a measure of short-term memory following a demonstration. In subsequent testing, 'C' had much greater long-term retention of number matrices than controls, especially for digits presented in her synaesthetic colours. Mills and

colleagues (2006) however selected case study 'MLS' on the basis of her synaesthesia alone, and still found that MLS remembered a greater number of novel Christian and surname pairings, as well as scoring much higher in tests of verbal ability (list-learning, the Rey Auditory –Verbal Learning Test) compared with matched controls. Considering the subjective reports of grapheme-colour synaesthetes, colour does seem to have a significant impact on memory (Mills et al., 2006; Smilek et al., 2002). As well as this, synaesthetes have reported the use of their colours as a strategy for learning on a daily basis (Luria, 1968; Rothen & Meier, 2010; Smilek et al., 2002; Yaro & Ward, 2007).

Exceptional memory ability in each of these cases was revealed using tests of synaesthesia-inducing material i.e. letters, words or digits. Reports have shown however, that this performance may be dependent on synaesthesia-inducing stimuli, such as in the case of C, who showed no memory advantage compared with matched controls for a series of simple line drawings (which did not elicit synaesthetic colours) [Smilek et al., 2002]. Similarly, the performance of MLS was comparable to that of controls in standardised visual tests that did not focus on grapheme-based stimuli (the Rey Complex Figure Test and the Benton Visual Retention test, revised) [Mills et al., 2006]. Enhanced verbal scores (e.g. Rey Auditory–Verbal Learning Test, Farnsworth–Munsell memory test) paired with ordinary visual scores (e.g. Rey Complex Figure recall) were similarly found in a group study by Yaro and Ward (2007), following the same pattern as in case study reports.

It must be considered that an exceptional memory advantage may simply not exist, but for a few isolated synaesthete cases, as proposed by Rothen and Meier (2009). They found no performance benefit linked to the synaesthetic colours elicited in a memory task involving black and coloured (congruent or incongruent with synaesthetic concurrents) digit matrices, either in immediate or delayed recall (Rothen & Meier, 2009). Following this, Rothen and Meier (2010) reported that group memory scores for synaesthetes were within ordinary ranges (one standard deviation) despite a performance advantage in episodic memory tests. This advantage was not found in tests of digit span or

visual short-term memory, following scores from the Wechsler Memory Scale (Revised edition) [WMS-R].

When analyzing the trends in the existing synaesthesia literature, one must address the essential differences that exist amongst grapheme-colour synaesthetes. While it is highly uncommon for two synaesthestes to report a given set of graphemes eliciting the same synaesthetic colours, categorical differences have been reported between so called ‘projector’ and ‘associator’ grapheme-colour synaesthetes. Projector synaesthetes are described as perceiving colours in eternal space, usually superimposed on the text they are reading (Smilek et al., 2001; Ward, & Mattingley, 2006). Associator synaesthetes however, experience colour in their ‘mind’s eye’ when they think of particular letters or words etc. (Dixon, Smilek, & Merikle, 2004; Ward & Mattingley, 2006).

Distinctive qualities may exist in the perceptual processing of projector and associator synaesthetes, which may offer a reason for the inconsistencies seen in the memory and attention literature mentioned (Dixon et al., 2004; Ramachandran & Hubbard, 2001b; Smilek et al., 2001; Edquist et al., 2006). Research has found projectors to be faster than associators at naming their synaesthetic colours (Dixon et al., 2004; Ward & Mattingley, 2006) and at detecting a target in a visual search task (Palmeri et al., 2002; Smilek et al., 2001; Smilek et al., 2003). However, some have found no significant difference in the performance of these subtypes (Edquist et al., 2006; Sagiv et al., 2006).

Considering the trends in the research thus far, the present study hypothesised that the experience of synaesthetic colours when presented with completely novel words would aid the encoding process upon learning (Cytowic, 2002; Paivio, 1969), and subsequently act as a cue for rapid and accurate recognition of letter sequences upon spelling recall from long-term memory stores (Smilek et al., 2002). Accordingly, synaesthetes would report using colours to aid the encoding of the non-words during the learning task. While previous research had assessed verbal ability in synaesthetes (Mills et

al., 2006), tested memory capacity using digit span tasks (Rothen & Meier, 2010), auditory learning tasks (Mills et al., 2006) and the recall of digit and colour matrices (Yaro & Ward, 2007), there has yet to be a study involving the learning and recall of completely novel words, recognising correct spellings.

Thus the present research assessed learning capacity for 27 novel, pronounceable non-words using a word recognition task involving the discrimination of these non-words as correctly or incorrectly spelled, amongst other unlearned words. Participants also completed a letter search task in which they searched for a target letter when presented with a number of common English words, thus accounting for the word superiority effect in the visual search paradigm (McClelland & Rumelhart, 1981). In both of these tasks it was predicted that synaesthetes would produce faster reaction times and make fewer errors than controls, and that these scores would each have a significant relationship with the number of unique colours perceived by the synaesthetes. Finally, participants completed a questionnaire to ascertain which learning strategies were most commonly used during the learning task, with the expectation that synaesthetes would primarily use their synaesthetic colours to aid memory.

METHOD

Design

This study employed a 'between-subjects' analysis within a quasi-experimental design, comparing the speed and accuracy of non-word recognition and target letter identification of grapheme-colour synaesthetes (both projectors and associators) to non-synaesthetic controls. A 'within-subjects' analysis was conducted to assess the relationship between the number of unique synaesthetic colours and the speed and accuracy of non-word recognition and letter search ability within the synaesthete group.

Participants

Nine grapheme-colour synaesthetes (6 female and 3 male; mean age: 19.9 years, SD: 1.9) and thirty-two non-synaesthetic controls (19 female and 13 male; mean age: 20.5 years, SD: 3.2) were recruited from the undergraduate student populations of Trinity College Dublin (TCD) and the Royal College of Surgeons in Ireland (RCSI) via email response to a poster and email campaign. Research credits alongside entry into a draw for a €50 voucher were offered for participation. All synaesthetes reported vivid colours for letters and entire words, and two experienced colours for spoken words, with five perceiving colours when a letter is spoken aloud. Eight reported seeing these colours in their 'mind's eye' (termed associators) and one experienced colours projected on letters and words when reading (projector) [Dixon et al., 2004], as well as in the mind's eye (ascertained via follow-up email). The majority of participants spoke English as a first language, with Spanish as the native language of one synaesthete (also fluent in English). None reported a positive assessment for dyslexia.

Materials

All tests were completed using 21.5" screen iMac desktop computers (1920 x 1080 pixels) with Mac OS X, version 10.5.8, with an accompanying iMac

Mighty Mouse and flat-key iMac keyboard. The stimuli were presented using Superlab 4.0 for Mac, each word displayed in black text, Calibri font and font size 80; written instructions were presented in the same colour and font, but were in font size 32 (see Appendix VIII for display example). All stimuli were presented on a white background to simulate normal reading conditions. Stimuli and the design of recognition and search tasks were purely visual in nature to avoid the interference of auditory stimuli, which may have elicited conflicting synaesthetic colours in those who perceived colours from sounds. All non-words were pronounceable, based on the most common phonemes in the English language and sourced from the online non-word generator website, http://www.soybomb.com/tricks/words/.

'The Synesthesia Battery' (Eagleman, Kagan, Nelson, Sagaram, & Sarma, 2007) was used via the Internet to assess for grapheme-colour synaesthesia (see www.synesthete.org for full details), with a set of printed instructions provided for participants (see Appendix XII). The information sheet/consent form, debriefing sheet, general questionnaire, instructions for the Synesthesia Battery and learning method questionnaire were printed in black, size 12, Arial font on white A4 paper (see Appendices IX-XI for full text). Participants were mostly tested in groups of two or more, seated away from one another in a quiet and well-ventilated computer laboratory which contained 14 identical computers, lit with overhead lights. Unfortunately, due to timetable constraints, two participants were tested alone, but the exact procedure was followed, counteracting any potential differences.

Pilot Study

A pilot study was first conducted (following the procedure below) using a male undergraduate student (age 22) to ensure all the task instructions were coherent and clear, and to highlight any programming errors in each section of the experiment.

The Synesthesia Battery – www.synesthete.org

The presence of synaesthesia was confirmed using 'The Synesthesia Battery' (Eagleman et al., 2007; accessed online at the above address). This involved a consistency test ('Grapheme Colour Picker Test') in which participants were asked to choose a colour for each of the 28 graphemes in the English alphabet, and consistency score evaluating the similarity of the colours was calculated automatically (a score lower than 1 is considered an indication of synaesthesia, with a score above 2 usually the result of a participant without synaesthesia). Following this, a 'Speed-Congruency Test' was completed, involving a Stroop-like task, which required participants to identify the graphemes in the correct or incorrect synaesthetic colour (a score between 85-100% indicating synaesthetic abilities). See www.synaesthete.org for further information.

Procedure

Once consent was confirmed, participants were given a questionnaire seeking demographic information and information related to grapheme-colour synaesthesia (for full question list see Appendix XI). The experiment consisted of a number of tasks, which included 1) the learning task, 2) a filler task, 3) a word recognition task, and 4) a letter search task.

1) Learning Task

Participants were presented with 27 non-words (nine each of five-letter, seven-letter and nine-letter words) in three-word blocks. Each three-word block was presented a total of three times, with each word presented alone at the centre of the screen for three seconds (following a blank screen for 600 milliseconds (ms) and the presentation of a plus symbol ('+') for 1s – see Appendix VIII for an example). To test for non-word retention, following each three-word block, participants engaged in a word-stem completion task, in which the first two/three letters of the word were provided alongside a

dialogue box for the participant to type in the remaining letters (see Appendix VIII for an example of screen presentation and Appendix IV for full non-word list and word-stem list). If participants answered correctly for each word-stem completion trial they would move onto the next three-word block. If one or more of the three non-words were misspelled, the last three words of the block would be presented again in the same manner as before and the test presented once more until all three non-words could be spelled correctly and long-term retention ensured. Participants repeated this process for each of the three letter non-word blocks. Participants were given a one-minute break and were asked not to talk. The instructions were given in red text on a white background and remained on the screen for the full minute.

2) Filler Task

To distract the participants and shift their focus before the word recognition task, participants engaged in a task identifying two-digit numbers as higher or lower in value than the preceding two-digit number. Each number was presented for two seconds following a blank screen (600ms) and a plus symbol ('+') in the centre of the screen (1s). To indicate that the present number was higher in value than the preceding number the participant was asked to press the 'x' key, if the number was lower they were to press the 'z' key. There were 50 two-digit numbers presented in random order overall. A trial version of the test was first implemented using a total of 5 trials to ensure the participants understood the response procedure. Participants were asked to complete each phase as quickly and accurately as possible (for full instructions and stimulus list see Appendix V).

3) Word Recognition Task

Participants then completed a word recognition task in which they were asked to identify 81 correctly or incorrectly spelled non-words, presented in random order. These included previously learned non-words (during 'the Learning Task'); the same non-words altered by one letter e.g. 'theek' changed to 'theel'; or unlearned non-words. A trial version of the test was first

implemented using a total of five trials to ensure the participants understood the response procedure. Reaction times for a response of ‘yes’ (press ‘z’) or ‘no’ (press ‘x’) were recorded upon the presentation of each non-word (which remained on the screen until a response was provided) following a blank screen (600ms) and a plus symbol (‘+’ for 1s). Participants were asked to complete each phase as quickly and accurately as possible (for full instructions and stimulus list see Appendix VI). Participants were given a one-minute break and were asked not to talk (instructions were given in red text on a white background and remained on the screen for the full minute).

4) Letter-search Task

To assess recognition speed for letter recognition in real words, participants were asked to search for a target letter in a series of common English words. The search for target letters ‘t’, ‘c’ and ‘s’ were assessed in three separate blocks of 20 words each. Participants were asked to indicate whether a target letter was present in the given word by pressing the ‘z’ key for yes or the ‘x’ key for no. Words were selected so that they did not contain the target letter of the previous trails to avoid false positives or confusion. Each word was presented in the centre of the screen one at a time, until a response was provided, following a blank screen (600ms) and a plus symbol (‘+’ for 1s). Participants were asked to complete each trial as quickly and accurately as possible (for full instructions and stimulus list see Appendix VII).

5) Learning Method Questionnaire

Participants were asked to indicate which method(s) they employed while completing the non-word Learning Task by ticking the appropriate box(es). The question was presented as follows:

Q What technique did you use to learn the non-words when they were presented to you?

Tick as appropriate:

( ) Saying the word aloud in your mind

( ) Noting the 'colour' of the overall word

( ) Noting the 'colour' of the individual letters

( ) Taking a 'mental picture' of the word

( ) Breaking the word into chunks or into its constituent syllables

( ) Focusing on the first or last letters

( ) Thinking of similar words

RESULTS

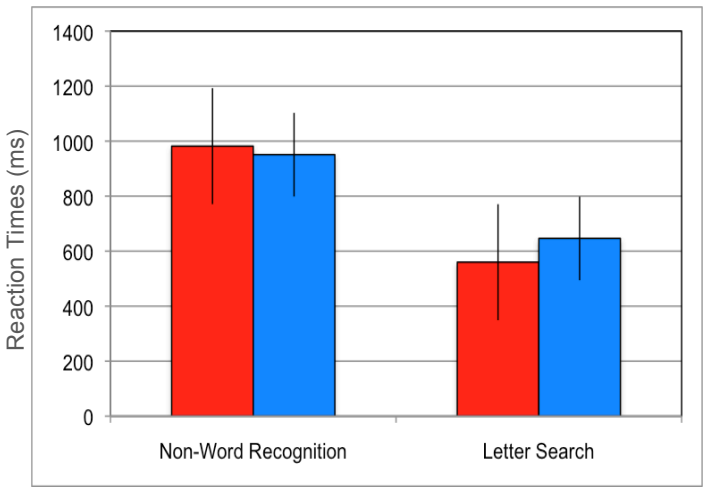

Figure 1. Means of central tendency scores (means or medians) for reaction times (ms) and accuracy scores (percentage correct) for grapheme-colour synaesthetes and controls in the non-word recognition task, including error bars (standard error).

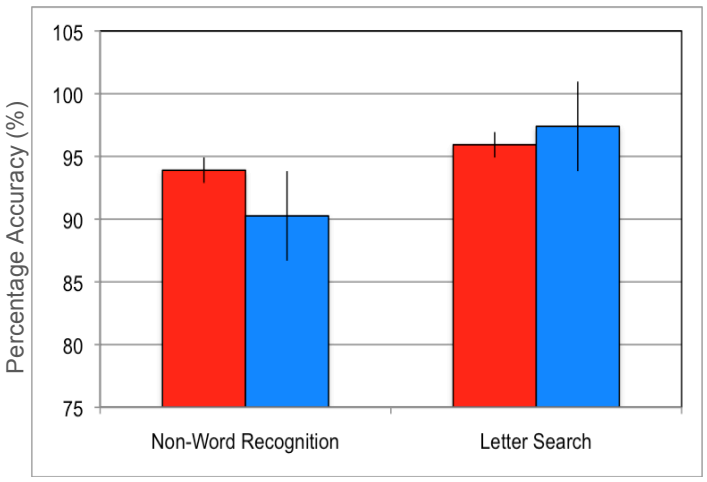

Figure 2. Means of central tendency scores (means or medians) for accuracy scores (percentage correct) for grapheme-colour synaesthetes and controls in the letter search task, including error bars (standard error).

The Synaesthesia Battery confirmed the presence of synaesthetic abilities in each of the synaesthete participants (mean: 0.65, SD: 0.33) [numerical score below 1 indicating synaesthesia], alongside a percentage accuracy of 85-100 percent (mean: 90.02%, SD: 2.71). Similarly, control participants were each screened with all receiving results greater than 2 (control mean: 2.25, SD: 0.55) or lacking variability in colour choice to determine a numerical value (within the ranges of expected scores for this test) [Eagleman et al., 2007].

The Statistical Package for the Social Sciences (SPSS, Version 18 for Mac) was used for all statistical analyses. It was hypothesised that grapheme-colour synaesthetes would produce more rapid and accurate responses in a test of non-word recognition (newly learned) and in a letter search task amongst real words (analyses (1), (2), (3) and (4)). In addition, it was predicted that the greater number of unique synaesthetic colours experienced by the synaesthetes, the more quickly and accurately correct responses would result in both of these tests (analyses (5), (6), (7) and (8)). All incorrect responses were removed from the data set. The remaining data for each participant was inspected to identify outliers using a boxplot, with means used as a measure of central tendency if less than five mild outliers (1.5 times the

Table 1. Summary of **Mann-Whitney U** analyses of reaction times (ms) and accuracy scores (percentage of correct responses) for grapheme-colour synaesthetes and controls in the non-word recognition task and the letter search task.

	U	z	p	Synaesthete Ranks (n = 9)	Control Ranks (n = 32)
Non-word recognition task					
(1) Reaction Times	128	-.5	NS	22.78	20.50
(2) Accuracy Scores	100	-1.4	NS	25.89	19.63
Letter Search Task					
(3) Reaction Times	90	-1.7	NS	15.00	22.69
(4) Accuracy Scores	115	-.94	NS	17.78	21.91

interquartile range) and no extreme outliers (3 times the interquartile range) were present, otherwise a median was used. The significance level was set at alpha = .05 for all statistical analyses. Figure 1. summarizes the group mean reaction times of the non-word recognition task and the letter search task for the synaesthete and control groups, while Figure 2. provides the accuracy scores (percentage of correct responses) for both of these tasks. Table 3. presents this information in tabular form (alongside standard deviations for each descriptive statistic) in Appendix II.

While a series of Levine's tests confirmed homogeneity of variance between the synaesthete group and the control group in comparisons of (1) non-word recognition reaction times; (2) non-word recognition accuracy scores; (3) letter search task reaction times; and (4) letter search accuracy scores, the results were found not to be normally distributed upon inspection of histograms and following a series of Shapiro-Wilk tests [the control group in analyses (1), (2), (3) and (4) produced significant results]. The data did not become normally distributed following a log transformation (base 10). Thus, a Mann-Whitney U test was implemented in each case, with the expectation that synaesthetes

Table 2. **Pearson's r** analyses of the correlation between number of unique synaestheteic colours and reaction times (ms) and accuracy scores (percentage of correct responses) for grapheme-colour synaesthetes in the non-word recognition task and the letter search task.

	r	n	p
Non-word recognition task			
(5) Unique Colours vs. Reaction Times	-.001	9	NS
(6) Unique Colours vs. Accuracy Scores	.277	9	NS
Letter Search Task			
(7) Unique Colours vs. Reaction Times	.009	9	NS
(8) Unique Colours vs. Accuracy Scores	.093	9	NS

(n = 9) would produce faster reaction times and fewer incorrect responses than controls (n = 32), thus produce lower ranks for synaesthesia in analyses (1) and (3), but higher ranks for synaesthetes in analyses (2) and (4) [see Table 1. for a full outline of results]. In each case the results revealed no significant differences between the groups, indicating no performance advantage for grapheme-colour synaesthetes in speed or accuracy of non-word recognition or letter search within real words with analyses (2) and (3) following the expected directions, but not (1) or (4).

A series of Pearson product-moment correlation coefficients were computed to assess the relationship between the number of unique synaesthetic colours experienced (judged by three independent raters, see Table 4. in Appendix III) by the synaesthetes and (5) the non-word recognition reaction times; (6) the non-word recognition accuracy scores; (7) the letter search task reaction times; and (8) the letter search accuracy scores of the synaesthetes. In each case there was no correlation or a very weak relationship between the two variables (see Table 2. for a full outline), with scatter plots revealing the widely dispersed nature of the data in each case. Scatter plots summarize these results in Figure 3., Figure 4., Figure 5., and Figure 6. for correlation analyses

(5), (6), (7) and (8) respectively. Overall, there was no observed relationship between the number of unique synaesthetic colours experienced by synaesthetes and the reaction times or accuracy scores in either task.

The learning method questionnaire identified that the majority of synaesthetes did not report the use of their synaesthetic colours as a memory aid during the learning task, with only two noting the distinctiveness of individual letters, and two using the colour of whole words to enhance memory. The most common strategy implemented by nearly all members of both the synaesthete group and the control group was ‘saying the word aloud in your mind’ (8/9 or 88.9% of synaesthetes; 30/32 or 93.75% of the controls).

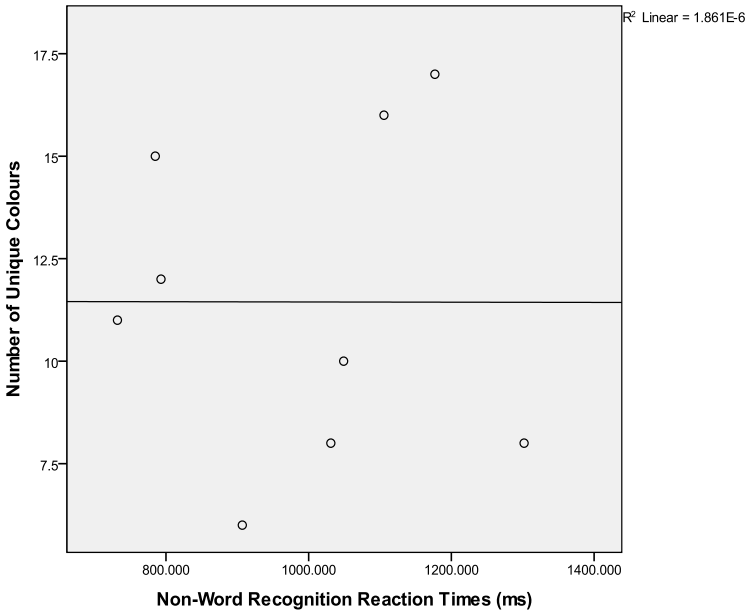

Figure 3. A scatter plot of no. of unique synaesthetic colours against non-word recognition reaction times (in milliseconds) for the synaesthete group (analysis 5).

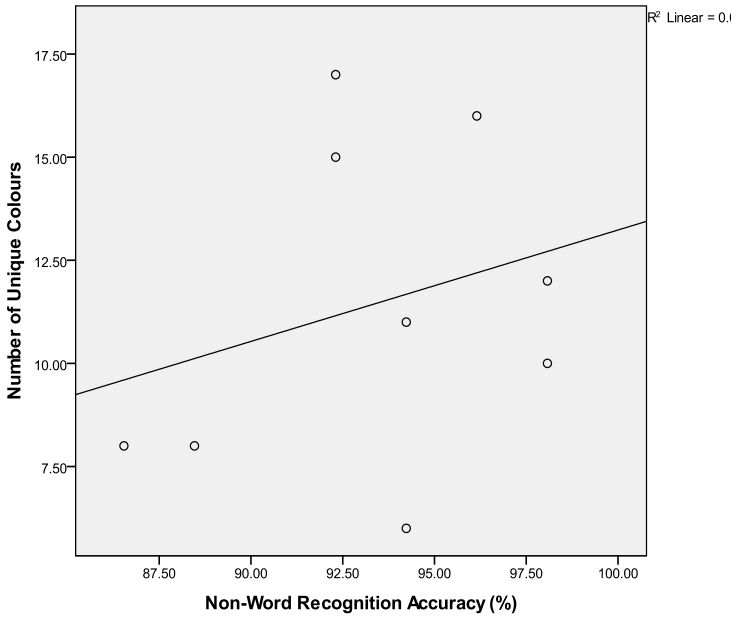

Figure 4. A scatter plot of no. of unique synaesthetic colours against non-word recognition accuracy scores (percentage correct) for the synaesthete group (analysis 6).

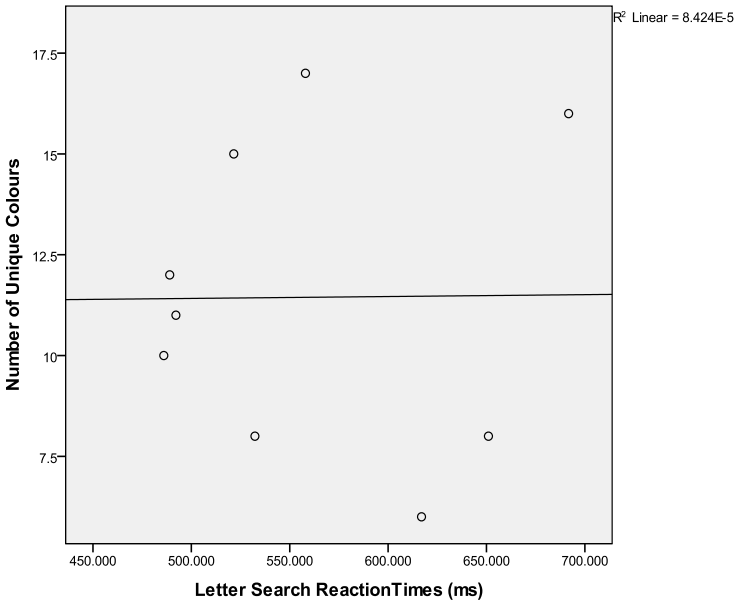

Figure 5. A scatter plot of no. of unique synaesthetic colours against letter search reaction times (in milliseconds) for the synaesthete group (analysis 7).

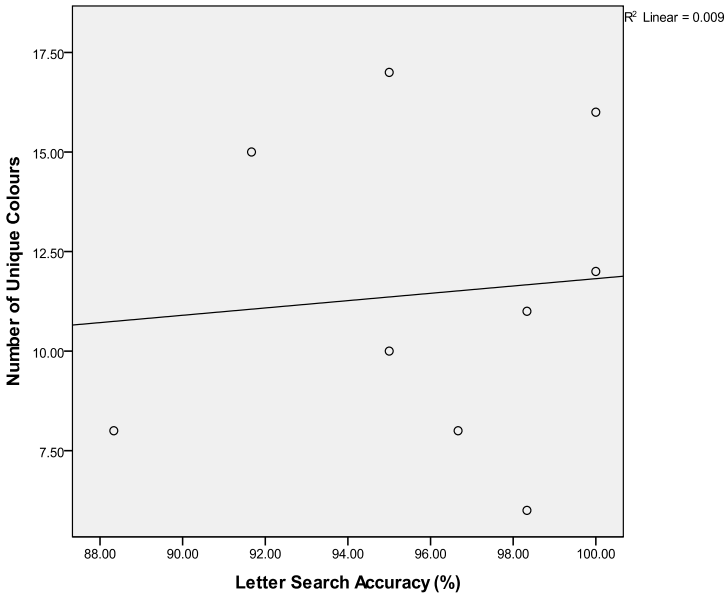

Figure 6. A scatter plot of no. of unique synaesthetic colours against letter search accuracy scores (percentage correct) for the synaesthete group (analysis 8).

DISCUSSION

The principle aims of this study were to investigate whether synaesthetic colours, which are induced during learning and recognition of new words, cause grapheme-colour synaesthetes to recognise correctly spelled words more quickly and accurately, and identify target letters more efficiently within words than a non-synaesthete control sample. It was similarly hypothesised that the greater number of synaesthetic colours experienced the faster the reaction times and accuracy of scores for synaesthetes in both of these tasks, and that synaesthetes would declare use of their synaesthetic colours to aid memory during the learning task. There were no observed significant differences between scores for grapheme-colour synaesthetes and controls for either the recognition task or the letter search task, and no relationship was found between the uniqueness of colours and any of these scores for the synaesthete group. As well as this, only two synaesthetes reported using their synaesthetic colours for letters and two using their perceived colours for whole words as memory aids during the learning task, with most implementing the same strategies as the control participants. The results indicate that synaesthesia does not have a significant impact on learning or word stimulus identification, which would distinguish these abilities from those of a normal population.

While the findings of this study contradict much of the case studies investigating the memory abilities of a single case grapheme-colour synaesthete, who each showed a strong influences of perceived colours alongside to-be-remembered digit matrices, first-name and surname pairs and in tests of verbal ability (Luria, 1968; Mills et al., 2006; Smilek et al., 2002), the findings are in line with Rothen and Meier (2009) who found no link between task performance and the presence of synaesthesia for both memory (memorizing digit matrices, scores on the Weschler Memory Scale-R) and performance in visual search tasks. Similarly, previous studies using the visual search paradigm had also found no difference between synaesthetes and matched controls (Edquist et al., 2006; Sagiv et al., 2006). There are a

number of robust reasons why the results in this case may show no significant differences from the performance of controls.

Amongst the statistical models used in this study, some were non-parametric in nature (due to the lack of normally distributed data). Models of this kind produce a less powerful analysis than their parametric equivalents, and thus may have failed to identify any significant results. While following the same sex bias as other studies (much greater number of females) [Barnett, Newell, Finucane, Corvin, Mitchell, 2008; Barnett & Newell, 2008; Baron-Cohen et al., 1996; Rich et al., 2005) and a similar synaesthete to control ratio [1:4] (1:13 Mills et al., 2006; 1:8 Smilek et al., 2001), the sample size was relatively small and the participants were from a specific age-bracket and educational status (university undergraduates) which may have rendered the participant sample unrepresentative of the general synaesthesia population.

The type of grapheme-colour synaesthetes tested must be also be addressed, as eight of the nine who participated in this study reported experiencing synaesthetic colours only in their 'mind's eye' (termed associators) and not overlaid on the text while reading (termed projectors) [Dixon et al., 2004; Smilek et al., 2001]. As previously outlined, projector synaesthetes have shown performance advantages in a synaesthetic colour naming (Dixon et al., 2004; Ward et al., 2006) and in letter search tasks (Palmeri et al., 2002; Smilek et al., 2001; Smilek et al., 2003). While synaesthetic colours arise during fewer tasks for associators (upon thinking of graphemes) compared with projectors (when reading, searching a visual scene or thinking), it may be postulated that the tasks presented in this study would reap poorer results for associators than projectors. The results advocate that associators are more likely to posses average memory abilities due to the nature of their synaesthesia.

These distinctive qualities in the perceptual processing of projector and associator synaesthetes may offer a reason for the inconsistencies seen in the memory and attention literature concerning synaesthesia, as many previous studies did not consistently account for such differences in the

course of data analysis (Dixon et al., 2004; Ramachandran & Hubbard, 2001b; Smilek et al., 2001; Edquist et al., 2006). It must also be considered that previous studies investigating memory were largely case study analyses which often sought the participation of a single individual following the discovery of their exceptional memory abilities (Luria, 1968; Smilek et al., 2002), whereas synaesthetes were selected on the basis of their synaesthesia alone in this study. Thus it may be reasoned that previous research proves misleading in some respects by raising expectations regarding memory ability in synaesthetes.

The instructions given during task implementation must be considered in addition, as the synaesthetes in this study were not explicitly told to use their synaesthetic colours to aid memory, but rather asked to comment on their methods once assessment had been completed. Synaesthetes may need focused attention to elicit their synaesthetic colours (Mattingley et al., 2006; Rich & Mattingley, 2003) and thus may require conscious engagement of this sort of attention during a learning task to reap any benefits of their distinctive colour patterns (Rothen & Meier, 2009).

Limitations and Future Research

Some limitations must be considered in the evaluation of this research. The tasks presented in this study were novel and experimental in nature, used due to the lack of a suitable standardised alternative. Perhaps the use of a standardised spelling and word recognition battery would provide normative and threshold scores to indicate exceptional spelling ability, and thus reveal if the synaesthetes tested remained within normal levels (as in Rothen and Meier, 2010) or if it was the case that the control sample were also exceptional in these abilities and confounding the findings.

Even though this study focused mainly on the learning and recognition of letter patterns within words, the presentation of non-words alongside a fabricated definition of the word would have potentially offered a more

ecologically valid context for non-word learning, as new words are always paired with meaning and context in everyday scenarios. If more time was available, this nuanced method might produce more ecologically valid results. Similarly, the repetition of each section of the experiment and the averaging of results would provide a more realistic pattern of results, as during this study reaction times for each stimulus were recorded only once.

As discussed, one must consider as well the small sample obtained and the bias that may have existed due to the imbalance of participants in the synaesthete (n = 9) and control (n = 32) groups. If more time allowed, a larger and more diverse sample of synaesthetes (more projectors) may have provided a more realistic sample of participants. The comparison of associators and projectors on a wider array of tasks is warranted for future research to identify robust differences as well as the replication of previous studies which did not account for these categorical differences amongst grapheme-colour synaesthetes. The impact of explicit instructions to use synaesthetic colours would be an interesting avenue for further research as it is speculated that focused attention is required for colours to be registered by the synaesthete later in cortical processing (Bargery, Barnett, Mitchell & Newell, 2009).

The focused use of synaesthesia may elicit a larger number of correct and more rapid responses compared with the use of average learning techniques, such as those taught in educational contexts to all students without regard for such abilities e.g. chunking to-be-learned materials, repetition, speaking words aloud etc. The study of the effect of the conscious versus the implicit use of synaesthetic colours as an aid for memory in everyday learning and spelling tasks may produce some interesting findings in addition to the current attention research in the synaesthesia literature.

CONCLUSIONS

Overall, the results indicated that the synaesthetic colours experienced by grapheme-colour synaesthetes may not have a significant effect on non-word recognition or letter search speed or accuracy rendering performances different from a non-synaesthetic control sample. In the same way, the results indicated that the number of unique synaesthetic colours for specific graphemes may not have a significant relationship with the speed or accuracy of scores in tests of word recognition or letter search abilities. Finally, synaesthetes may not use their synaesthesia as a learning aid unless specifically instructed to do so, mostly use the same techniques as non-synaesthetes during learning tasks involving graphemes. The conflict regarding synaesthesia and the memory and recognition components of spelling ability is still unresolved. Further replication of results in larger group studies and the comparison of scores across different types of grapheme-colour synaesthetes may highlight the most essential pieces of the puzzle yet.

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APPENDICES

APPENDIX I: Letter of ethical approval from the Psychology Research Ethics Committee in Trinity College, Dublin.

COLÁISTE NA TRÍONÓIDE, BAILE ÁTHA CLIATH | TRINITY COLLEGE DUBLIN
Ollscoil Átha Cliath | The University of Dublin

30th November 2010

F.A.O. Suzy Percy

School of Psychology Research Ethics Committee

Dear Suzy,

I am pleased to inform you that your application entitled "Influence of colour-grapheme synaethesia on spelling recall/recognition" has been approved by the School of Psychology Research Ethics Committee.

Yours sincerely,

Dr. Tim Trimble
Chair
School of Psychology Research Ethics Committee

SCHOOL OF PSYCHOLOGY
Aras an Phairsaigh
Trinity College
Dublin 2

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Dámh na nEolaíochtaí Sóisialta agus Daonna,
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APPENDIX II:

Table 3. Means and standard deviations for central tendency scores of synaesthetes and controls for the recognition task and the letter search task.

Table 3. Means and standard deviations of reaction times (ms) and accuracy scores (percentage correct) for grapheme-colour synaesthetes and controls for the non-word recognition task and for the letter search task).

		Reaction Times		Percentage Accuracy	
Group	n	Mean	SD	Mean	SD
Recognition Task					
Synaesthetes	9	986.81	195.22	93.99	3.98
Controls	32	950.78	205.03	90.26	6.2
Letter Search Task					
Synaesthetes	9	559.86	76.04	95.93	3.92
Controls	32	646.41	154.13	97.4	2.39

APPENDIX III:

Table 4. The number of unique colours experienced by the synaesthete group

Table 4. The number of unique synaesthetic colours expereinced for the grapheme-colour synaesthesia group amongst the 68 letters in the English alphabet (Latin-based), as judged by three independent raters.

	Number of Unique Synaesthetic Colours								
Participant	1	2	3	4	5	6	7	8	9
No. of colours (out of a possible 26)	6	8	11	12	15	10	16	19	8

APPENDIX IV: 1) Learning Task list of experimental word stimuli and task instructions.

Instructions to participants:

Slide 1
You will now be presented with a series of non-words in 3-word blocks.
Learn each word carefully as you will be tested after each block.
Left CLICK THE MOUSE to begin.

Slide 2 (word-stem completion instructions, presented again if one or more of the word-stems were completed incorrectly)
You will now be presented with the first few letters of the words you have just learned.
Please type in the remaining letters.
If you enter the wrong letters for one or more of these trials you will see the same block of words again and will repeat the test.

Slide 3 (presented if all word-stem completions correct)
You will now proceed to the next block
Left CLICK THE MOUSE to continue

Non-word stimuli and word-stems presented during the Learning Task.

Non-word stimuli	Word-stem	Non-word stimuli	Word-stem
chayt	ch_____	mations	mat_____
thoob	th_____	antlood	ant_____
hithe	hi_____	curangs	cur_____
cheen	ch_____	farmals	far_____
trude	tr_____	boltinted	bolt_____
bowce	bo_____	rabirtgan	rab_____
soych	so_____	shuseness	shu_____
foize	fo_____	pestorded	pest_____
theek	th_____	refeative	ref_____
lootton	loo_____	aurammuth	aur_____
poffolo	pof_____	sorsioned	sor_____
janizes	jan_____	monsifica	mon_____
steugem	ste_____	banglated	ban_____
catechy	cat_____		

APPENDIX V: 2) Filler Task instructions.

Instructions to participants:

Slide 1
You will be presented with 2-digit numbers, one after another.
Please indicate whether each number is HIGHER or LOWER than the number before it.
Press the SPACEBAR to continue.

Slide 2
Press 'z' for LOWER or 'x' for HIGHER
Please respond as quickly and accurately as possible
Press the SPACEBAR to continue.

Slide 3
Place your index fingers over the 'z' and 'x' now
Press either key for the first number
Press the SPACEBAR to begin.

Participants were presented with 89 two-digit numbers between 10 and 99.

e.g. 13, 45, 67, 89, 54, 47, 34, 24, 14, 35, 56, 33, 78, 88, 70, 57, 47, 48, 29, 11.

APPENDIX VI: 3) Non-Word Recognition Task list of experimental stimuli and task instructions.

Instructions to participants:

Slide 1
You will now be presented with a sequence of words.
These will be words you have previously learned OR slight variations of these words OR completely new words.
For each word presentation you must indicate whether you learned the word in the previous task or not.
Press the SPACEBAR to continue

Slide 2
Trail Run
You will now complete a trial run of this task
Type 'z' for NO or 'x' for YES to indicate if you previously learned each word
Press the SPACEBAR to continue

Slide 3
Please respond as quickly and accurately as possible
Place your index fingers over the 'z' and 'x' now
Press the SPACEBAR to begin.

Slide 4
Testing will now start
Remember to respond as quickly and accurately as possible.
Place your index fingers over the 'z' and 'x' now
Press the SPACEBAR to begin.

Words used in trial run:

foime (altered spelling)
chayt (previously learned non-word)
werdly (unlearned non-word)
fardals (altered spelling)
mistanded (unlearned non-word)

Non-words and variations presented during the Non-Word Recognition Task.

Original Non-Words	Non-Word Variations	New Non-Words
chayt	chayf	dayke
thoob	hitne	favel
hithe	trube	gitow
soych	creen	culch
trude	theel	scall
bowce	poftolo	operson
cheen	steuyem	knorier
foize	matioks	judilow
theek	cumangs	pitural
lootton	bolfinted	rapheme
poffolo	snuseness	reguesent
janizes	reteative	helverson
steugem	sorsioted	flakenage
catechy		morporate
mations		
antlood		
curangs		
farmals		
boltinted		
rabirtgan		
shuseness		
pestorded		
refeative		
aurammuth		
sorsioned		
monsifica		
banglated		

52 words presented in total

APPENDIX VII: 4) Letter Search Task experimental word stimuli and task instructions.

Instructions for participants:
(Example provided is for search of letter 't'. The same instructions were repeated for searches of the letter 'c' and 's')

Slide 1
You will be presented with a series of real words, one after another.
Please indicate whether the words contain the letter: 't'
Press the SPACEBAR to continue

Slide 2
Press 'z' for YES or 'x' for NO if a letter containing 't' is presented.
Please respond as quickly and accurately as possible.
Press the SPACEBAR to continue

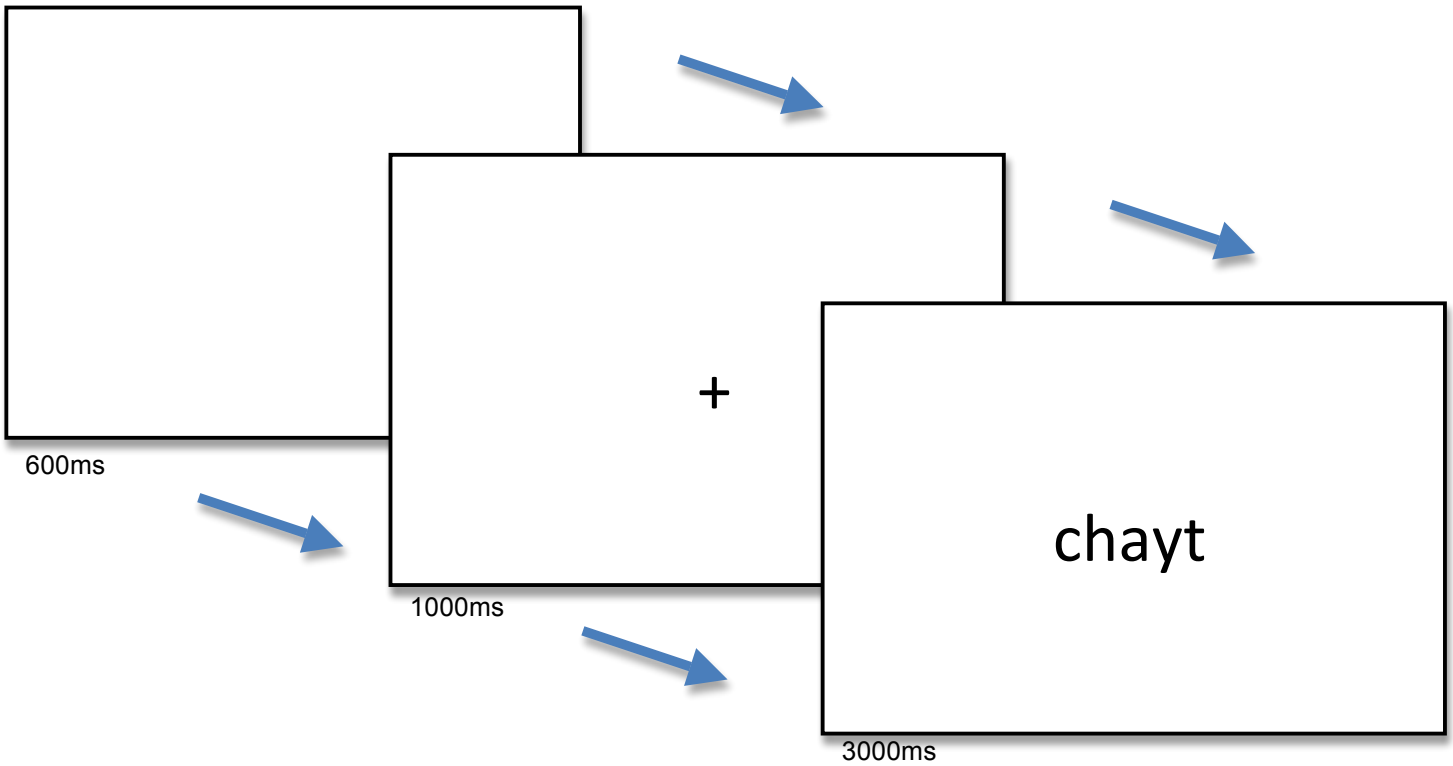
Slide 3
Place your index fingers over the 'z' and 'x' now
Press the SPACEBAR to begin.

Common English words presented during the Letter Search Task.

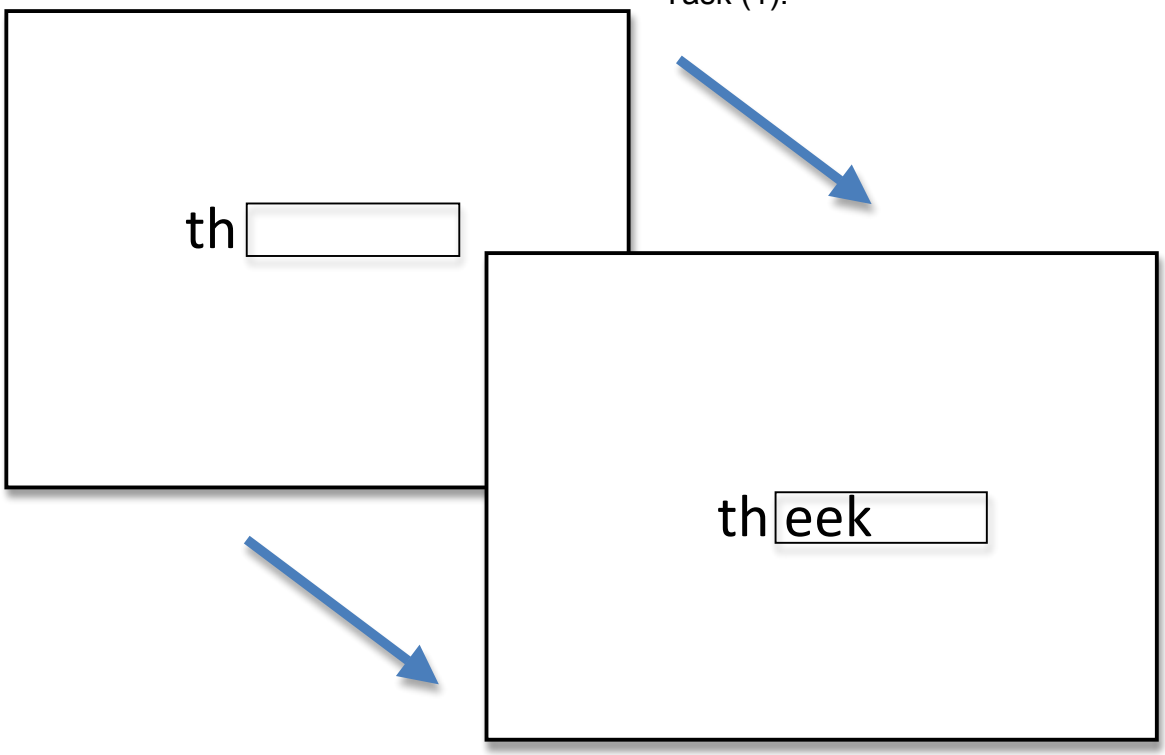
Block 1	**Block 2**	**Block 3**
Contains letter 't'	**Contains letter 'c'**	**Contains letter 's'**
telephone	cable	survey
thirst	carnival	silent
ticked	creature	seven
trench	cream	shook
table	certain	stack
store	scream	browse
industry	machine	verse
ventral	facial	friendship
recent	block	revise
steel	become	morsel
Does not contain 't'	**Does not contain 'c'**	**Does not contain 's'**
grand	grounding	dealer
opened	money	rhyme
floor	indigo	energy
kicked	lounge	yellow
downside	believe	dozen
swirl	plough	bargain
horse	grumble	mobile
broke	shining	liquid
fresh	razor	idiom
learning	mouse	entire

APPENDIX VIII: Example of stimulus presentation.

Stimulus presentations (with times in milliseconds) for all experimental tasks (1, 2, 3, 4).



Word-stem completion presentations during the testing phase of the Learning Task (1).



APPENDIX IX: Information sheet/Consent form.

Information sheet

My name is Suzy Percy and I am a 4th year undergraduate student studying psychology in Trinity College Dublin. The following study aims to assess memory capacity and spelling ability. You will be asked to complete a number of learning and recall tasks, which will each be followed by two-minute break. To assess for the presence of grapheme-colour synaesthesia you must provide a user name and password for the online testing website www.synesthete.org. Your results will be kept strictly private and only you and I, the researcher, will have access to your results.

All data collected will be coded and be kept strictly confidential and anonymous in accordance with the Data Protection Acts of 1988 and 2003 and your identity will not be divulged in any written report. The information you provide will be coded and analysed as part of a larger data set.

Data will be kept securely and be retained for a period of ten years in accordance with the policy of the School of Psychology, Trinity College Dublin. You also have to right to access any findings of the study in accordance with the Freedom of Information Act 1997 and 2003.

Your participation in the study is voluntary and you are free to withdraw from it at any time without prejudice.

If you require any further information please do not hesitate to contact me or my supervisor, Prof. Fiona Newell:

Suzy Percy (Researcher)
c/o School of Psychology,
Áras an Phiarsaigh,
Trinity College,
Dublin 2.
E-mail: percys@tcd.ie
Tel: 086 3687071

Prof. Fiona Newell (Supervisor)
TCIN Lloyd Institute,
Trinity College Dublin,
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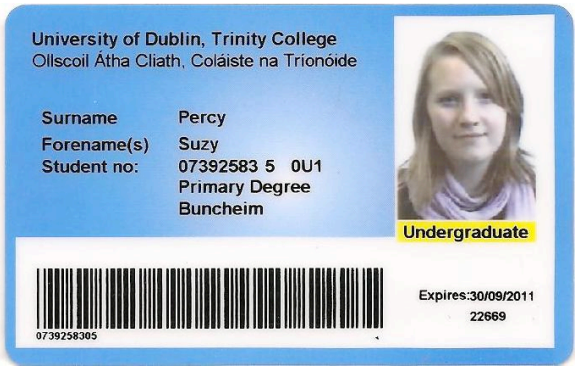
Email: fiona.newell@tcd.ie
Tel: 01 896 8402



I have read the above information and give my consent to participate in this research study:

Print Name: _____

Signed: _____ Date: _____

APPENDIX X: Debriefing Sheet

Debriefing Sheet

Thank you very much for participating in this study. The purpose of this research is to compare the spelling ability and memory span of grapheme-colour synaesthetes with a (non-synaesthetic) control group.

It must be revealed that the number-reaction test was used as a filler task to distract you from the newly learned words in the learning phase, and not to assess reaction speeds as stated. This deception was necessary to ensure recognised words in the testing phase were recalled from the long-term memory and not held temporarily in iconic memory stores.

The data you have provided will be kept strictly confidential and anonymous in accordance with the Data Protection Acts of (1988, 2003) and stored securely for 10 years in line with the Freedom of Information Act 2003.

If you have any questions about the study or wish to see a copy of the final report please do not hesitate to contact either myself or my supervisor (see details below):

Suzy Percy (Researcher)
c/o School of Psychology,
Áras an Phiarsaigh,
Trinity College,
Dublin 2.
E-mail: percys@tcd.ie
Tel: 086 3687071

Prof. Fiona Newell (Supervisor)
TCIN Lloyd Institute,
Trinity College Dublin,
Dublin 2.
Email: fiona.newell@tcd.ie
Tel: 01 896 8402

Futher Information:

If you would like to learn more about synaesthesia or if this study has raised any concerns regarding spelling or learning, information about various organisations are listed below:

Synaesthesia	Spelling and Learning difficulties
TCD Multisensory Cognition Group www.tcd.ie/Neuroscience/multisensory **The Synesthesia Battery (test used)** www.synesthete.org **U.K. Synaesthesia Association** www.uksynaesthesia.com **American Synesthesia Association** www.synesthesia.info **Synesthesia Resource Centre** (Articles, documentaries and videos) www.bluecatsandchartreusekittens.com	**TCD Centre for Practice and Student Learning** www.tcd.ie/CAPSL (01) 896 3601 **TCD Disability Service** www.tcd.ie/dyability (01) 896 3111 Room 2054, Arts Building, Trinity College Dublin. **Dyslexia Association of Ireland** www.dyslexia.ie (01) 6790276 Suffolk Chambers 1 Suffolk Street Dublin 2, Ireland. **National Council for Guidance in Education** www.ncge.ie National Centre for Guidance in Education, 1st Floor 42/43, Prussia Street, Dublin 7, Ireland. (01) 8690715/6

APPENDIX XI: Demographic information and synaesthesia specific questionnaire.

Please answer the following questions:

1) Gender: _____

2) Age:_____

3) Highest level of education completed/or currently studying (please tick):

( ) Primary school

( ) Junior Certificate

( ) Leaving Certificate

( ) Vocational course/Apprenticeship

( ) University Undergraduate Degree

( ) University Postgraduate course (MA, PhD)

4) Occupation:
_____

4) Is English your first/native language?

( ) Yes

( ) No. My first native language is _____.

5) Please list all secondary languages and indicate proficiency in brackets
(fluent, conversational, beginner) [e.g. French (beginner)]

_____

_____

7) How would you rate your spelling ability on a scale of 1-7?
(1 = very poor, ---------- 4 = average, ----------7 = exceptional/excellent)

_____

8) Have you been formally diagnosed with Dyslexia? (Yes/No/Don't know) _____

Questions regarding grapheme-colour synaesthesia

9) Do you see colours when you READ individual letters? ( ) Yes ( ) No

10) Do you see colours when you READ written words? ( ) Yes ( ) No

10 a) What has the LARGEST influence on the OVERALL COLOUR of a word?
(A single letter, vowels, first letter, last letter, word meaning etc.)

( ) N/A

_____

_____

_____

10 b) Do letters or words have a TEXTURE in addition to a specific colour?

( ) N/A

_____

_____

_____

11) Do you see colours when you HEAR spoken words? ( ) Yes ( ) No

12) Do you see colours when you HEAR individual letters of the alphabet? ( ) Yes ( ) No

13) Additional comments:

_____

_____

_____

APPENDIX XII: Synaesthesia battery online user instructions (www.synaesthete.org).

Phase 1: The Synesthesia Battery

CAREFULLY READ ALL INSTRUCTIONS – not all parts of the online test need to be filled out.

1. Complete the registration form and click register. It may be easiest to use your college email and password. You will be able to access your results using these log in details after the study if you want to.

2. Enter your details to log in.

3. Enter the details of the researcher:

   Researcher's Email Address: **percys@tcd.ie**
   Researcher's Name: **Suzy Percy**

4. Read the information letter. Tick the box in the lower left corner. Click Proceed.

5. Scroll down and **only tick the second box 'Letters->Color' (you do not need to answer the other questions**). Scroll to the bottom of the page and click 'Next step'

6. The test will take approx 10 minutes. Click 'Go to Tests'

7. Follow the instructions on screen.

   N.B. If you do not see synaesthetic colours, pick the most appropriate colour from the selection or choose the 'no color' option.

8. When both tests are completed, scroll to the bottom of the screen and click 'Submit' (**ignore the other questions**).